INCH-POUND
MIL-M-38510/174A
AMENDMENT 4
23 APRIL 1999

SUPERSEDING
AMENDMENT 3
04 OCTOBER 1988

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, INVERTER, NAND BUFFER DRIVER,
STROBED HEX INVERTER/BUFFER, VOLTAGE LEVEL SHIFTER,
MONOLITHIC, SILICON, POSITIVE LOGIC

Inactive for new design after 9 August 1996

This amendment forms a part of MIL-M-38510/174A, dated 30 April 1984,
and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.3, add new case outlines as follows:

"N F-5A (16-lead, .440" x .285" x .115"), flat package"
"T F-2A (14-lead, .390" x .260" x .115"), flat package".

PAGE 4

4.4.1, add new paragraph as follows:

"f. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance."

PAGE 6

* TABLE I, input leakage current test, (I_{IH}), max limits column, device types 01 and 03, delete "45" and substitute "100"; (I_{IL}), max limits column, device types 01 and 03, delete "-45" and substitute "-100".

TABLE I, input capacitance, device type and limit columns, delete and substitute the following:

"

01	---	12.0
02,03,04		7.5

"

PAGE 7

TABLE I, 3-state disable delay time, conditions column: Delete "R_L = 200 kΩ" and substitute "R_L = 1.0 kΩ.

PAGE 9

TABLE II, add "10,11" on line 8 for class B device. Delete line 12 in its entirety.

FSC 5962

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:44Z
Check the source to verify that this is the current version before use.

PAGE 10

FIGURE 1: Delete “cases A, C, D, X, and Y” and substitute “cases A, C, D, T, X, and Y” for device type 01. Delete “cases E, F, and Z” and substitute “cases E, F, N, and Z” for device type 03.

PAGE 13

FIGURE 4, note 3: Delete first sentence; add “Except for V_{DD} and V_{SS}, terminals shall be connected through resistors whose value is 2 kΩ to 47 kΩ.”

PAGE 14

FIGURE 5, note 1: Delete first sentence and substitute: “Except for V_{DD} and V_{SS}, terminals shall be connected through resistors whose value is 2 kΩ to 47 kΩ.”

PAGE 15

FIGURE 6: Delete “$t_{TLH(1)}$” and $t_{THL(1)}$” on input generator and substitute “t_r and $t_f \leq 20$ ns”.

PAGES 19 AND 20

FIGURE 6, test circuit, output load resistor: Delete “R_L = 10 kΩ ±5 %” and substitute “R_L = 1.0 kΩ ±5 %”.

PAGES 21 THROUGH 27

TABLE III: Delete “cases A, C, D, X, and Y” and substitute “cases A, C, D, T, X, and Y” for device type 01.

PAGE 23

* TABLE III, test number 63, subgroup 1, max limit column: Delete “6” and substitute “600”.

* TABLE III, test numbers 64 through 69: subgroup 1, max limit column: Delete “1.0” and substitute “100”; subgroup 2, max limit column: Delete “45” and substitute “100”.

* TABLE III, test number 70, subgroup 1, max limit column: Delete “-6” and substitute “-600”.

PAGE 23 AND 24

* TABLE III, test numbers 71 through 76, subgroup 1, max limit column: Delete “-1.0” and substitute “-100”; subgroup 2, max limit column: Delete “-45” and substitute “-100”.

PAGE 25

TABLE III, test numbers 101 through 106, maximum limit column: Delete “7.5” and substitute “12.0”.

PAGES 33 THROUGH 46

TABLE III: Delete “cases E, F, and Z” and substitute “cases E, F, N, and Z” for device type 03.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:44Z
Check the source to verify that this is the current version before use.

PAGE 38

* TABLE III, test number 125, subgroup 1, max limit column: Delete “8” and substitute “800”.

* TABLE III, test numbers 126 through 133: subgroup 1, max limit column: Delete “1.0” and substitute “100”; subgroup 2, max limit column: Delete “45” and substitute “100”.

* TABLE III, test number 134, subgroup 1, max limit column: Delete “-8” and substitute “-800”.

PAGE 39

* TABLE III, test numbers 135 through 142, subgroup 1, max limit column: Delete “-1.0” and substitute “-100”; subgroup 2, max limit column: Delete “-45” and substitute “-100”.

* TABLE III, test number 149, subgroup 3, max limit column: Delete “0.54” and substitute “0.64”.

PAGE 53

TABLE III, note 5/: Delete “10 kΩ ±5 %” and substitute “1.0 kΩ ±5 %”.

PAGE 54

4.4.3: Delete paragraph 4.4.3c in its entirety.

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Navy - AS, CG, MC, OS, SH
Army - AR, MI, SM
Air Force - 19, 99

(Project 5962-1845)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:44Z
Check the source to verify that this is the current version before use.